

PERSONAL DETAILS

- Full name: Patrick Michael Maher
- Location: Brisbane, Australia
- Nationality: Australian
- Languages: English (native)
- ORCID: 0009-0001-2320-4626

EDUCATIONAL QUALIFICA- TIONS & ACADEMIC AWARDS

University of Queensland

Doctor of Philosophy (PhD) in Statistics

Brisbane, Australia

Jan 2026 – Present

- Advisors: Prof. Andrew Martin and Prof. James McGree
- Research focus: Variational Methods in Clinical Trials

University of Queensland

Master of Biostatistics

Brisbane, Australia

Feb 2024 – Jun 2026

- GPA: 6.625/7.0
- Advisors: Dr. Ding Ma and Prof. Andrew Martin
- Research thesis title: “Bayesian Semiparametric AFT Joint Modelling for HrQoL and PFS”
- Dean’s Commendation for Academic Excellence, Semesters 1 and 2 2025

University of Queensland

Bachelor of Mathematics (Statistics)

Brisbane, Australia

Feb 2021 – Nov 2023

- GPA: 5.8/7.0
- Dean’s Commendation for Academic Excellence, Semester 2 2023

PROFESSIONAL AFFILIATIONS & MEMBERSHIPS

- 2024 – Present: Queensland Branch Early Career Representative, Statistical Society of Australia
- 2024 – Present: Member, Australian Clinical Trials Alliance: Statistics in Trials Interest Group (ACTA-STInG)
- 2025 – Present: Student Member, International Society for Clinical Biostatistics
- 2025 – Present: Student Member, International Biometric Society
- 2026 – Present: Student Member, International Society for Bayesian Analysis

EMPLOYMENT HISTORY

- Jan 2024 – Present Senior Research Assistant, UQCCR, University of Queensland
 - Prepared analysis plans, derived variables, and programmed statistical analyses in R, HPC, Stan and Python.
 - Performed sample size and adaptive design calculations across multiple endpoints within clinical trials.
 - Implemented complex hierarchical, joint, multivariable and multivariate models primarily within the Bayesian paradigm
 - Constructed semi-parametric Bayesian methods and optimisation techniques (Variational Bayes, Expectation Propagation, Integrated Nested Laplace Approximations)
 - Authored written reports; presented statistical methodology findings at national (and international) conferences.

OTHER
RESEARCH
EXPERIENCE

- Lead statistician on the *PIVO (Reducing blood culture contamination with the use of a needle-less blood draw device (PIVO Pro): An adaptive group sequential randomized controlled trial)* trial. Working with clinicians for design of statistical analysis, endpoints, data management and randomisation.
- Lead statistician on the *ASITI-T1D (A phase 1, randomised, double-blind, placebo-controlled, multicentre, single-dose and multi-dose escalation study to investigate the safety, and pharmacodynamics of subcutaneously administered proinsulin peptide/calcitriol liposomes (ASITI-201) in adults with stage 3 Type 1 Diabetes.)* trial. Working with clinicians for design of statistical analysis, endpoints, data management and randomisation. Derived novel Bayesian adaptive dose-finding methodology leveraging prior elicitation of expert belief.
- Contribution in statistical design considerations for *PROACTIVE* and *ANVU* adaptive trials. With *PROACTIVE* as a Bayesian adaptive hierarchical trial design within endometriosis. And the *ANVU* study eliciting prior information in a Bayesian design for gynaecological cancer.
- Acknowledged in the “Acknowledgements” section of Takefala TG, Mayr HL, Doola R, Johnston HE, Hodgkinson PJ, Andelkovic M, Macdonald GA & Hickman IJ. *Implementing early feeding after liver transplant using implementation frameworks: a multimethod study. Nutrition in Clinical Practice.* 2024;1–15
- Meta analysis work for a systematic review investigating the global standard of peripheral intravenous catheter (PIVC) care in acute hospital settings as measured against healthcare quality indicators/outcomes, with contribution to analyses and manuscript writing with planned submission in December 2024
- Mentoring and assisting PhD and Masters students within statistical analyses for their research. Also mentoring junior statisticians in clinical trial design/analysis including but not limited to Bayesian methods
- Chaired, organised and participated within University of Queensland’s Trial Statistician (UQTS) monthly meetings, with the intention of sharing knowledge and utilising expertise within University of Queensland for trials
- Presented, prepared and facilitated workshop on efficient Bayesian trial design. Showcased variational bayesian techniques in “From MAMS to Bayesian trial design” workshop during the ULTRA showcase for a group of approximately 30 Masters and PhD-level statisticians.
- Presented to BEAT-CALCI lead investigators about Bayesian concepts involving “Hierarchical modelling and information sharing” as well as “Rapid approximate inference in complex Bayesian trial designs”, showcasing key Bayesian trial concepts as well as innovative techniques such as Variational Bayes and Laplace Approximations to a non-technical audience.

Journal Articles

1. Mayr H., Murray E., **Mahe P.**, Hayes L., Wang W., Kelly J., Palmer M., Martin A, Hickman I. (Accepted *Sep 2025*) "Evaluating clinician sustainment of a Mediterranean-style dietary pattern approach in routine care for coronary heart disease and type 2 diabetes utilising Normalisation MeASURE Development (NoMAD) instrument," *Nutrition in Clinical Practice*.
Contribution: Performed main and sub analyses, devised appropriate outcomes, assisted drafting manuscript, created plots and data visualisations.
2. Ma D., **Mahe P.**, Martin A., (Re-submitted *June 2026*) "Joint Modelling Using Semiparametric Accelerated Failure Time Approaches," *Biometrical Journal*. arXiv:2606.16080.
Contribution: Derived Bayesian modelling, conducted simulations, wrote manuscript content, employed various semi-parametric approaches.
3. Lovegrove J., Shinnors L., Ray-Barruel G., **Mahe P.**, Mihala G., Smith S., McDonnell M., Fritts J., Williams E., Ven S., Vico da Silva J., Ozen N., Richard C. (Submitted *Mar 2026*) "Peripheral intravenous catheters in acute hospitals: A systematic review and meta-analysis of global care quality," *International Journal for Quality in Health Care*.
Contribution: Performed primary, secondary and sensitivity analyses, constructed plots, assisted drafting manuscript, ascertained proper statistical structure with modern methods.

Presentations

1. **Mahe P.**, Martin A. (November 2025) "Quick Estimates in Missing Data Structures with Variational Bayes," presented at Australian Trials Methodology (AusTriM) Conference, Melbourne, Australia. (Poster, speaker)
2. Mayr L., Hayes L., Wang W., Murray E., Kelly J., **Mahe P.**, Martin A., Palmer M., Hickman I. (October 2025) "From knowledge to action and sustainment: Translating an evidence-based dietary pattern approach into routine care in a public health service," presented at the Evidence and Implementation Summit, Melbourne, Australia. (Poster)
3. **Mahe P.**, Martin A. (October 2025) "Quick Bayesian Approximations for Clinical Trials," presented at 'Through our Eyes', Brisbane, Australia. (Poster, speaker)
4. Ma D., **Mahe P.**, Martin A. (August 2025) "Joint Modelling Using Semiparametric Accelerated Failure Time Approaches: Application to Health-Related Quality of Life Analysis," presented at International Society for Clinical Biostatistics Conference, Basel, Switzerland. (Oral)
5. Ma D., **Mahe P.**, Martin A. (August 2025) "A Bayesian Semiparametric Joint Modelling Framework for Non-Proportional Hazards," presented in Australian Trials Methodology (AusTriM) Trial Analysis Special Interest Group Meeting (Invited, Online, Oral, speaker)
6. **Mahe P.**, Thomas R., Martin A. (December 2024) "ASITI-201-T1D: A Randomized Dose-Finding Trial," presented at Australian Clinical Trials Alliance Conference, Melbourne, Australia. (Poster)
7. **Mahe P.**, Martin A. (December 2024) "Incorporating Frequentist Group Sequential Methods in Bayesian Trial Design," presented at Australian Clinical Trials Alliance Conference, Melbourne, Australia. (Oral, speaker)

ACADEMIC
REFEREES

- **Professor Andrew Martin**
Professor of Innovative Clinical Trials, UQCCR, University of Queensland
Herston QLD 4006, Australia
Email: andrew.martin@uq.edu.au
Role: Supervisor (January 2024- Present)
- **Dr Ding Ma**
Postdoctoral Research Fellow, UQCCR, University of Queensland
Herston QLD 4006, Australia
Email: ding.ma@uq.edu.au
Role: Co-worker and research collaborator (March 2024- Present)